Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 2)

1. Suppose $X_1, \ldots, X_n$ is a random sample from an $\mathsf{Bernoulli}(p)$ distribution. Laplace's estimator of the proportion p is

$$\hat{p}_L = \frac{(\sum_{j=1}^n X_j) + 1}{n+2}.$$

   (a) What is the sampling distribution of $S = \sum_{j=1}^n X_j$

   Solution. As $X_1, \ldots, X_n \sim \mathsf{Bernoulli}(p)$, $S \sim \mathsf{Binomial}(n, p)$. For the questions that follow, it will be useful to recall $\mathbb{E}_p(S) = np$ and $\mathrm{Var}_p(S) = np(1-p)$

   (b) Is $\hat{p}_L$ an unbiased estimator of p? If not, determine the bias of $\hat{p}_L$.

   Solution. Applying the definition of bias,

$$\mathrm{bias}(\hat{p}_L; p) = \mathbb{E}_p\left(\frac{S+1}{n+2}\right) - p = \frac{\mathbb{E}_p(S)+1}{n+2} - p = \frac{np+1}{n+2} - p = \frac{1-2p}{n+2}$$

   The bias of $\hat{p}_L$ is zero only for $p = \frac{1}{2}$.

   (c) Determine the mean-squared error of $\hat{p}_L$.

   Solution. Since

$$\mathrm{Var}_p(\hat{p}_L) = \mathrm{Var}_p\left(\frac{S+1}{n+2}\right) = \frac{\mathrm{Var}_p(S)}{(n+2)^2} = \frac{np(1-p)}{(n+2)^2},$$

   the mean-squared error is

$$\mathrm{MSE}(\hat{p}_L; p) = \mathrm{Var}(\hat{p}_L) + \mathrm{Bias}(\hat{p}_L)^2 = \frac{np(1-p) + (1-2p)^2}{(n+2)^2}.$$

   (d) For which values of p does $\hat{p}_L$ have smaller MSE than the usual estimator $\hat{p} = n^{-1}\sum_{j=1}^n X_j$.

   Solution. The usual estimator $\hat{p} = S/n$ is unbiased and has

$$\mathrm{MSE}(\hat{p}) = \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

   Therefore $\hat{p}_L$ has smaller MSE when

$$\frac{np(1-p) + (1-2p)^2}{(n+2)^2} < \frac{p(1-p)}{n}.$$

   This is equivalent to

$$n(1-2p)^2 < 4(n+1)p(1-p).$$

   or

$$p^2 - p + \frac{n}{4(2n+1)} < 0.$$

   The above quadratic has a global minimum at $\frac{1}{2}$ and roots at

$$\frac{1 \pm \sqrt{(n+1)/(2n+1)}}{2}.$$

   Thus $\hat{p}_L$ has smaller MSE for

$$\frac{1 - \sqrt{(n+1)/(2n+1)}}{2} < p < \frac{1 + \sqrt{(n+1)/(2n+1)}}{2}.$$

2. Suppose $X_1, \ldots, X_n$ is a random sample from an $\mathsf{Exponential}(\lambda)$ distribution. Here $\lambda > 0$ is the rate parameter so the pdf is $f(x;\lambda) = \lambda \exp(-\lambda x)$ for $x \geq 0$ and $f(x;\lambda) = 0$ for $x < 0$. It may be helpful to know the following: If $Y \sim \mathsf{Gamma}(\alpha, \beta)$, then

$$\mathbb{E}(Y) = \frac{\alpha}{\beta}, \quad \mathrm{Var}(Y) = \frac{\alpha}{\beta^2}, \quad \mathbb{E}(Y^{-1}) = \frac{\beta}{\alpha - 1}, \quad \mathrm{Var}(Y^{-1}) = \frac{\beta^2}{(\alpha-1)^2(\alpha-2)}$$

and the moment generating function of Y is

$$M_Y(t) = \left(\frac{\beta}{\beta - t}\right)^{\alpha}, \text{ for } t < \beta.$$

The $\mathsf{Exponential}(\lambda)$ distribution coincides with the $\mathsf{Gamma}(1, \lambda)$ distribution.

(a) Show the sampling distribution of $\bar{X}$ is $\mathsf{Gamma}(n, n\lambda)$ $(\star)$.

Solution. The moment generating function of $\bar{X}$ is

$$M_{\bar{X}}(t) = \mathbb{E}\left(\exp(\bar{X}t)\right) = \mathbb{E}\left(\prod_{j=1}^{n} \exp(X_j t/n)\right).$$

As $X_1, \ldots, X_n$ are independent,

$$M_{\bar{X}}(t) = \prod_{j=1}^{n} \mathbb{E}\left(\exp(X_j t/n)\right) = \prod_{j=1}^{n} M_{X_j}(t/n).$$

Using the form of the moment generating function for the exponential distribution

$$M_{\bar{X}}(t) = \prod_{j=1}^{n} \left(\frac{\lambda}{\lambda - (t/n)}\right) = \left(\frac{\lambda}{\lambda - (t/n)}\right)^n = \left(\frac{n\lambda}{n\lambda - t}\right)^n$$

which is the moment generating function of a $\mathsf{Gamma}(n, n\lambda)$ distribution. As moment generating functions are unique, the distribution of $\bar{X} \sim \mathsf{Gamma}(n, n\lambda)$.

(b) Determine the method of moments estimator for λ.

Solution. Using the given moments of the gamma distribution, $\mathbb{E}_\lambda(X) = 1/\lambda$. Equating $1/\lambda$ with the first sample moment $\bar{X}$, we obtain the method of moments estimator $\hat{\lambda} = 1/\bar{X}$.

(c) Determine the bias of the estimator from part (b).

Solution. Applying the definition of bias,

$$\mathrm{bias}(\hat{\lambda}; \lambda) = \mathbb{E}_\lambda\left(1/\bar{X}\right) - \lambda = \frac{n\lambda}{n-1} - \lambda = \frac{\lambda}{n-1},$$

where we have used the fact that $\bar{X} \sim \mathsf{Gamma}(n, n\lambda)$ and the given moments of the gamma distribution.

(d) Determine the constant c such that $c/\bar{X}$ is an unbiased estimator of λ.

Solution. For the estimator $c/\bar{X}$ to be unbiased for λ we need $\mathbb{E}_\lambda(c/\bar{X}) = \lambda$. Again using the fact that $\bar{X} \sim \mathsf{Gamma}(n, n\lambda)$ and the given moments of the gamma distribution,

$$\mathbb{E}_\lambda\left(c/\bar{X}\right) = c\,\mathbb{E}_\lambda\left(1/\bar{X}\right) = \frac{cn\lambda}{n-1}.$$

Upon taking $c = (n-1)/n$, we see $\mathbb{E}_\lambda(c/\bar{X}) = \lambda$, and hence it is an unbiased estimator of λ.

(e) Does the estimator from part (d) have a smaller MSE than the method of moments estimator? Justify your answer.

Solution. Note the variance of $1/\bar{X}$ exists only when $n > 2$. As the estimator in part (d) is unbiased,

$$\text{MSE}\left(c/\bar{X};\lambda\right) = \text{Var}\left(c/\bar{X}\right) = c^2\text{Var}\left(1/\bar{X}\right),$$

and

$$\text{MSE}\left(\hat{\lambda};\lambda\right) = \text{Var}\left(1/\bar{X}\right) + \text{bias}(\hat{\lambda};\lambda)^2.$$

As $c < 1$ and $\text{bias}(\hat{\lambda};\lambda)^2 \geq 0$, we see that the estimator in part (d) has a smaller MSE than the method of moments estimator.

3. Suppose $X_1, \ldots, X_{2n+1}$ is a random sample. Our statistical model is $\mathcal{F} = \{f(x;\theta) = g(x-\theta); \theta \in \mathbb{R}\}$, where g is some fixed pdf that is symmetric about 0 and satisfying

$$\int_{-\infty}^{\infty} |x| g(x) dx < \infty.$$

Using results from ordered statistics (STAT2003 - Claim 46) it can be shown that the sample median M has sampling distribution with pdf

$$f_M(x;\theta) = \frac{(2n+1)!}{n!\,n!}[F(x;\theta)]^n[1-F(x;\theta)]^n f(x;\theta), \qquad x \in \mathbb{R},$$

where $F(x;\theta)$ is the cdf for $f(x;\theta)$.

(a) Show that $F(\theta + x;\theta) = 1 - F(\theta - x;\theta)$, for $x \in \mathbb{R}$.

Solution. Each pdf $f(x;\theta)$ in the model $\mathcal{F}$ is symmetric about θ since g is symmetric about 0. This implies $F(\theta;\theta) = 1/2$. For $x \geq 0$,

$$F(\theta+x;\theta) - F(\theta;\theta) = \int_{\theta}^{\theta+x} f(u;\theta)\,du = \int_{\theta}^{\theta+x} g(u-\theta)\,du = \int_0^x g(v)dv,$$

where we used the substitution $v = u - \theta$. Also,

$$F(\theta;\theta) - F(\theta-x;\theta) = \int_{\theta-x}^{\theta} f(u;\theta)\,du = \int_{\theta-x}^{\theta} g(u-\theta)\,dv = \int_{-x}^{0} g(v)\,dv$$

where we again used the substitution $v = u - \theta$. As g is symmetric about 0,

$$\int_{-x}^{0} g(v)\,dv = \int_0^x g(v)dv.$$

Hence

$$F(\theta+x;\theta) - \frac{1}{2} = \frac{1}{2} - F(\theta-x;\theta),$$

which gives

$$F(\theta+x;\theta) = 1 - F(\theta-x;\theta).$$

The same identity follows for $x < 0$ by replacing x with $-x$.

(b) Show that M is an unbiased estimator of θ.

Solution. From part (a), the density of M is

$$f_M(x;\theta) = \frac{(2n+1)!}{n!\,n!}[F(x;\theta)]^n[1-F(x;\theta)]^n f(x;\theta), \qquad x \in \mathbb{R},$$

Using symmetry of f and the identity from part (b),

$$\begin{aligned} f_M(\theta + x) &= \frac{(2n+1)!}{n!\,n!}[F(\theta + x;\theta)]^n[1 - F(\theta + x;\theta)]^n f(\theta + x;\theta) \\ &= \frac{(2n+1)!}{n!\,n!}[1 - F(\theta - x;\theta)]^n[F(\theta - x;\theta)]^n f(\theta - x;\theta) \\ &= f_M(\theta - x;\theta). \end{aligned}$$

Thus the sampling distribution of M is symmetric about θ. Therefore, provided the expectation exists, $\mathbb{E}_\theta(M) = \theta$ and M is an unbiased estimator of θ. It remains to check that the expectation exists. As $0 \le F(x;\theta) \le 1$,

$$\begin{aligned} \int_{-\infty}^{\infty} |x| f_M(x)dx &\le \frac{(2n+1)!}{n!\,n!}\int_{-\infty}^{\infty} |x| f(x;\theta)dx = \frac{(2n+1)!}{n!\,n!}\int_{-\infty}^{\infty} |x| g(x-\theta)dx \\ &\le \frac{(2n+1)!}{n!\,n!}\int_{-\infty}^{\infty} |u+\theta| g(u)du \\ &\le \frac{(2n+1)!}{n!\,n!}\left(\int_{-\infty}^{\infty} |u| g(u)du + |\theta|\int_{-\infty}^{\infty} g(u)du\right) < \infty. \end{aligned}$$

Hence , the expectation exists.

4. Suppose $X_1, \ldots, X_n$ is a random sample from a Geometric(p) distribution, whose support is the positive integers $\{1, 2, \ldots\}$. As $\mathbb{E}_p(X) = 1/p$, the method of moments estimator is $\hat{p} = 1/\bar{X}$.

   (a) Is $\hat{p}$ a consistent estimator of p? Justify your answer.

   Solution. Since the X_i are independent and identically distributed with finite mean and variance, the law of large numbers can be applied to conclude $\bar{X}_n \xrightarrow{p} 1/p$. The function $h(x) = 1/x$ is continuous on $(0, \infty)$. We can apply the continuous mapping theorem to conclude $h(\bar{X}_n) = 1/\bar{X}_n \xrightarrow{p} h(1/p) = p$. Hence, $1/\bar{X}_n$ is a consistent estimator of p.

   (b) Carry out a small simulation study to investigate the finite sample properties of $\hat{p}$. Choose a value of p in $(0, 1)$. For each sample size $\{10, 50, 100, 500, 1000\}$, generate $B = 10,000$ independent samples from the Geometric(p) distribution, and for each of these samples compute $\hat{p}$. For each value of n, use the results of the simulation to estimate the bias, variance, and MSE of $\hat{p}$. Scale these quantities by n, (ie calculate $n\times$bias, $n\times$variance, and $n\times$MSE) and present the results in the table. What do you notice?
   Note: The `R` function `rgeom` simulates from the geometric distribution supported on $\{0, 1, 2, \ldots\}$. To generate from the geometric distribution on $\{1, 2, \ldots\}$ you will need to add one to the simulated values.

   Solution.

```
p <- 0.75
B <- 10000
ns <- c(10, 50, 100, 500, 1000, 5000)

results <- sapply(ns, function(n) {
    phat <- replicate(B, 1 / mean(rgeom(n,p)+1))
    c(nBias = n*(mean(phat)-p),
      nVar  = n*var(phat),
      nMSE  = n*mean((phat - p)^2))
})

results
            [,1]       [,2]       [,3]       [,4]       [,5]        [,6]
nBias 0.1884538 0.2120617 0.1550797 0.1711054 0.1299458 0.06691654
nVar  0.1292716 0.1388154 0.1405989 0.1401587 0.1415751 0.14128482
nMSE  0.1328102 0.1397009 0.1408253 0.1402033 0.1415778 0.14127158
```

When $p = 0.75$ and the sample size is n, the variance of the estimator $\hat{p}$ is roughly $0.14/n$ and the bias is roughly $0.2/n$. The main contribution to the MSE is from the variance. This makes sense since the MSE can be written as the sum of variance and squared bias.

5. The negative binomial distribution is used to model over-dispersed counts, that is counts where the variance is larger than the mean. We write $\mathsf{NB}(r, p)$ for the distribution with probability mass function

$$f(x) = \binom{x + r - 1}{r - 1} p^r (1 - p)^x, \quad x = 0, 1, 2, \ldots,$$

where $r \in (0, \infty)$ and $p \in (0, 1]$. If $X \sim \mathsf{NB}(r, p)$, then

$$\mathbb{E}(X) = \frac{r(1-p)}{p} \quad \text{and} \quad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}.$$

Let $X_1, \ldots, X_n$ be a random sample from $\mathsf{NB}(r, p)$.

(a) Determine the method of moments estimator for (r, p).

Solution. Write $\bar{X}$ for the sample mean as usual and

$$\hat{\sigma}^2 = n^{-1} \sum_{j=1}^{n} X_j^2 - \bar{X}^2.$$

Then the method of moments estimator satisfies

$$\bar{X} = \frac{r(1-p)}{p} \qquad \text{and} \qquad \hat{\sigma}^2 = \frac{r(1-p)}{p^2}.$$

We can solve for p as

$$\hat{\sigma}^2 = \frac{r(1-p)}{p} \cdot \frac{1}{p} = \bar{X} \frac{1}{p}.$$

So the method of moments estimator for p is $\hat{p} = \bar{X}/\hat{\sigma}^2$. We can now solve for r as

$$\bar{X} = \frac{r(1-p)}{p} = \frac{r(1 - \bar{X}/\hat{\sigma}^2)}{\bar{X}/\hat{\sigma}^2} = \frac{r(\hat{\sigma}^2 - \bar{X})}{\bar{X}}$$

So the method of moments estimator for r is $\hat{r} = \bar{X}^2/(\hat{\sigma}^2 - \bar{X})$.

(b) Under what conditions does your method of moments estimator belong to the parameter space.

Solution. We want to know when $\hat{r} > 0$ and $\hat{p} \in (0, 1]$. We have $\hat{p} \in (0, 1]$ whenever

$$0 < \frac{\bar{X}}{\hat{\sigma}^2} \leq 1.$$

So $\hat{p} \in (0, 1]$ whenever $\bar{X} \leq \hat{\sigma}^2$. Similarly, $\hat{r} \in (0, \infty)$ whenever

$$0 < \frac{\bar{X}^2}{\hat{\sigma}^2 - \bar{X}} < \infty.$$

So $\hat{r} \in (0, \infty)$ whenever $\hat{\sigma}^2 > \bar{X}$. The method of moments estimator of (r, p) belongs to the parameter space provided $\hat{\sigma}^2 > \bar{X}$.